

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5.	(a)			[Dark lines are formed when] atoms / gases in the star's atmosphere (1) absorb light (1) accept absorb colour at particular wavelengths (accept different frequencies) (1) [identifying the element]	3			3		
	(b)			Cosmic microwave background radiation, accept CMBR (1) is detected throughout the Universe / <u>gamma rays</u> have been stretched [to longer wavelength] (1) [Cosmological] red shift (1) shows that light (wavelengths) is stretched / galaxies are moving away from us (1) Accept shows that the Universe starts from a single point or the Universe is expanding for either but award each point only once	4			4		
				Question 5 total	7	0	0	7	0	0